Energy Bench: a live, gate-aligned benchmark of day-ahead forecasts for European power markets

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Methodology preprint, 28 October 2026
energybench.foretec.co · github.com/ujohn33/foretec-energy-bench

Abstract

Comparisons of energy forecasting models usually rest on backtests, which are easy to get subtly wrong: the information available when a forecast would have been sold is replaced by whatever is convenient to download later. Energy Bench runs every entry live instead. Each day at 11:30 CET/CEST, thirty minutes before the day-ahead (SDAC) gate closes, every model forecasts the next day's price, wind and solar generation and load on the 15-minute grid for six bidding zones (BE, NL, FR, DE-LU, DK1, DK2). Unlike other live benchmarks, Energy Bench runs the models itself, on one information set that it freezes at the cut-off and publishes, including three numerical weather prediction models, so foundation models are compared with and without the same weather covariates under a real market deadline. Errors are pooled per series against a seasonal-naive baseline, and models are ranked by a Bradley–Terry rating on the Elo scale with a day-bootstrap interval. The field spans baselines, linear and gradient-boosted models on the frozen weather, eighteen time-series foundation models from ten families, ensembles, and the grid operators' own forecasts. Over the first twenty-four live days, covariates roughly halve the error of foundation models: on matched races the seven that accept covariates natively cut their error by 45–50% (58–69% on wind, 58–63% on solar), and zero-shot they beat a LightGBM model trained on the same covariates. Model size, from 3M to 331M parameters, matters far less, and median ensembles of the covariate models lead the standings. The grid operators' forecasts remain the bar: level with the best entries on wind before the gate, ahead on solar and load, and ahead on both after it. A model enters the benchmark by pull request.

1 Introduction

Day-ahead forecasts of price, renewable generation and load are the inputs of every bidding decision on European power markets, and claims about how well a given model makes them are rarely comparable. Three things go wrong in the usual backtest. First, the information set drifts: a forecast nominally made at 11:30 is built from weather runs that were published hours later, from actuals that were still provisional, or from a TSO forecast that only appears after the gate. Second, the target drifts away from the market's: hourly values where the market now trades quarter-hours, generation definitions that differ between data providers, actuals that were later revised. Third, nothing stops an author from choosing the window, the baseline or the metric after seeing the results.

A fourth gap has opened with time-series foundation models. In operational energy forecasting, exogenous inputs are the standard, not an option: wind and solar forecasts are built on numerical weather predictions, load forecasts on temperature and calendar effects, and day-ahead price forecasts on forecasts of load, renewable infeed and fuel cost [8–10]. The grid operators' own forecasts, the industry reference, are NWP-driven, and the energy forecasting competitions supplied weather forecasts

as a matter of course [5]. The benchmarks on which foundation models are compared are, by contrast, mostly univariate. A model that forecasts tomorrow's solar output from the last four weeks of solar output alone is being tested on a task no practitioner would set it, and a ranking built that way says little about which model to deploy.

A live benchmark that hands every entry the same inputs closes these holes by construction. If the model runs at the real cut-off, on the data really available, weather forecasts included, and its forecast is locked before the real gate, there is no look-ahead to argue about, and the record accumulates one day at a time in public. The price is patience: a live record is short and its standings are noisy on any given day. The design below takes that noise seriously rather than hiding it: errors are pooled before they are normalised, ranks come with intervals, and only live forecasts are reported.

Energy Bench went live on 2 October 2026 with three zones (BE, NL, FR) and three targets; total load was added on 6 October and three zones (DE-LU, DK1, DK2) on 15 October. The code and the configuration file that define the benchmark are versioned in a public repository. This paper describes that configuration as of 27 October 2026 and the results to date.

2 Related work

Static benchmarks. Forecasting competitions have shaped the field since the M-competitions [1]; the Monash archive [2], GIFT-Eval [3] and fev-bench [4] standardise the evaluation of general-purpose and foundation time-series models on fixed datasets. In energy, the Global Energy Forecasting Competitions [5] posed realistic tasks as one-off events, and Lago et al. [10] published an open benchmark for day-ahead price forecasting with fixed splits and day-ahead exogenous forecasts for five markets. Fixed datasets have two weaknesses for the present purpose. They describe a past regime, and energy systems change quickly; and their test periods may have been seen, directly or through pretraining corpora, by the models they evaluate. Retrospective energy studies also often leave the information set implicit: several price-forecasting studies use ENTSO-E's day-ahead wind and solar forecasts as inputs although those are published only after 18:00 on $D-1$, six hours after the auction closes [7].

Covariates. The value of weather inputs in energy forecasting is long established. Wind and solar power forecasting has been NWP-based since its beginnings, and the wind and solar tracks of GEFCom2014 gave every team ECMWF forecasts [5]; load forecasting rests on temperature, and the reviews of price forecasting single out forecasts of load and renewable generation as the most informative exogenous drivers [8–10]. Foundation models have only recently gained the means to use such inputs: Chronos-2, TimesFM, TiRex-2, Timestral and the t0 models accept known-future covariates, by different mechanisms and with different success. Whether and how much they benefit is an empirical question that a univariate benchmark cannot answer, and it is the question an operational user needs answered first.

Live benchmarks. Two recent platforms evaluate forecasts made before the outcome exists, and each leaves open a question that Energy Bench is built to answer.

TS-Arena [6] implements a forecast pre-registration protocol for time-series foundation models: a round hands out each series' history up to its last published value and closes registration before the next value is recorded. Its 14 challenges draw on SMARD (German day-ahead price, load and generation), Gridstatus (US ISO hub prices, load and generation) and Fingrid (Finland), at 15-minute resolution with a one-day horizon and hourly with a three-day horizon, from a 1,000-step context. Models are scored by MASE and ranked by an online Elo rating whose interval comes from replaying the rounds in random order. Two features limit what its rankings say about energy forecasting. *First, there are no covariates*: the foundation models forecast wind, solar and load from the series' own history, without the weather forecast that any operational forecaster of those quantities would use, so the benchmark measures a task no market participant faces, and a model's ability to exploit covariates, which in our results halves the error of the same models (Section 4), is not measured at all. *Second, the information set is not tied*

to a market gate: each round starts at the last published value, so the window follows each source's publication delay rather than the market. A German price forecast registered on day X starts after the delivery day that cleared on X and covers a later day than the next auction needs, and the generation and load windows do not align with a delivery day at all. What a forecaster could have known at a decision time is therefore not defined, and no forecast corresponds to a bid that could have been placed.

Energy-Arena [7] is an open, API-based platform whose challenges are defined by their market deadline: forecasts are submitted before 12:00 CET on $D-1$ for the whole next day, as point (RMSE), quantile (weighted interval score) or ensemble (CRPS) forecasts, with leaderboards over rolling windows. It is gate-aligned, but it compares *submissions*, not models under common conditions. Participants run their own models on any inputs they have, including proprietary weather, and describe them only if they choose to; they may also keep their forecasts hidden. Since the information set cannot be verified, a difference on the leaderboard may reflect better data rather than a better method, and entries cannot be reproduced. The platform names no reference models, so whether foundation models are represented at all depends on who submits; and its wind, solar and load challenges cover only Austria and Germany-Luxembourg (day-ahead prices extend to thirteen further zones). In industry, the Predico platform [14] scores live renewable forecasts submitted to Elia and 50Hertz against a real gate, for enrolled forecasters and without publishing their inputs.

Positioning. Energy Bench takes Energy-Arena's operational framing (the day-ahead gate defines the task) and TS-Arena's pre-registration logic (only a forecast made before the outcome exists is free of leakage), and closes the gaps between them by running every model itself. Each day it fetches one information set at the gate-aligned cut-off, freezes and publishes it, and gives it to every entry: the information set is enforced rather than declared, and any forecast can be reproduced from its archived inputs. Foundation models from six families also run with the same frozen covariates, so the value of the weather is measured model by model rather than assumed away; all four targets are forecast in six zones; and the grid operators' own forecasts, before and after the gate, serve as references. The price is that entries are submitted as code rather than as forecasts, and proprietary data cannot be used. Table 1 summarises the differences.

3 The benchmark

This section defines the benchmark as it runs: the forecasting task, the daily timeline and what a model may know at the cut-off, where the actuals come from, the covariates frozen for every entry, the field of entries, and how forecasts are scored, ranked and valued.

3.1 Task

For each zone $z \in \{\text{BE}, \text{NL}, \text{FR}, \text{DE-LU}, \text{DK1}, \text{DK2}\}$ and target $y \in \{\text{price}, \text{wind}, \text{solar}, \text{load}\}$ (24 series), a forecast made on issue day $D-1$ covers every quarter-hour of delivery day D in local time: 96 values, or 92 and 100 on clock-change days. Price is the SDAC day-ahead price (EUR/MWh); wind is onshore plus offshore generation, solar is solar generation and load is total load (MW). Models return a point forecast and optionally the nine deciles $q_{0.1}, \ldots, q_{0.9}$. Danish load is published only hourly, so DK load forecasts are scored on hourly averages of forecast and actual: detail that the actual cannot resolve is neither rewarded nor penalised.

3.2 Timeline and information set

The benchmark day is defined by three clock times in CET/CEST (Figure 1). At the *cut-off*, 11:30 on $D-1$, a snapshot of everything public is taken and every model runs on it. The *gate* is 12:00, the SDAC gate closure: a live run that finishes after it is marked late and never scored, and a day's forecasts are locked once written: the record consists of the forecasts as they were at the gate. Prices are scored on $D-1$ once the auction for D has cleared (14:45 and 20:45 runs). Wind, solar and load are scored at

Table 1: Live energy forecasting benchmarks compared, as described in their papers and on their websites (October 2026).

	TS-Arena [6]	Energy-Arena [7]	Energy Bench (this paper)
Task timing	from each series' last published value; no market gate	submission before the 12:00 day-ahead gate, whole next day	11:30 cut-off, locked before the 12:00 gate, whole next day
Zones and targets	14 challenges (SMARD, Gridstatus, Fingrid)	wind, solar, load: AT, DE-LU; price: 15 zones	price, wind, solar, load in each of BE, NL, FR, DE-LU, DK1, DK2
Who runs models	participants (containers)	participants (API submissions)	the benchmark, from submitted code
Information set	series history only	anything before the gate; self-declared, unverified	frozen and published: history, three NWP models, load forecast, fuel
Covariates for foundation models	none	if a participant adds them	ten foundation models also with the same covariates
Foundation models	the focus of the field	none named; depends on submissions	ten families, and ensembles
Transparency	platform code public	method description and forecasts optional	code, inputs and forecasts public
Scores	MASE	RMSE, WIS, CRPS	MAE, RMSE, bias, pinball; relative to seasonal naive, pooled per series
Ranking	online Elo, replay bootstrap	per metric over rolling windows	Bradley–Terry Elo, day bootstrap
References	statistical baselines	optional benchmark models	baselines; the TSOs' forecasts before and after the gate

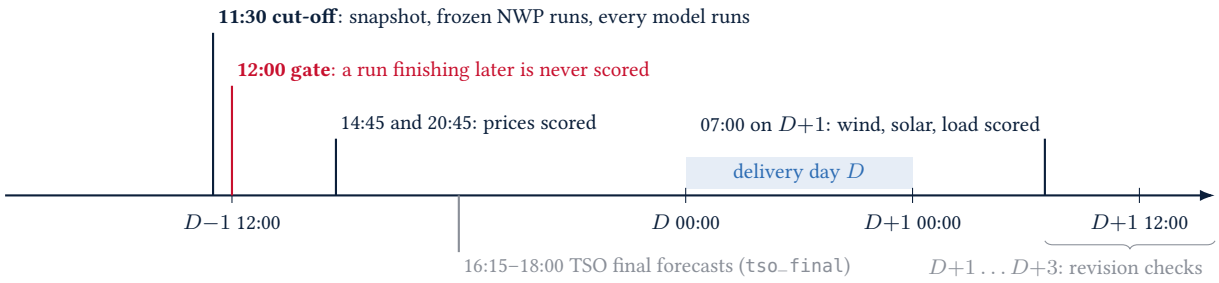


Figure 1: One benchmark day (not to scale). Everything a model sees is fixed at the 11:30 cut-off; the forecast must be locked before the 12:00 gate. The TSOs' final day-ahead wind and solar forecasts appear only after the gate and enter as references.

07:00 on $D + 1$ if every quarter-hour of D has been published, otherwise at the next scoring run once 95% has.

What a model may know at the cut-off is set per target. Prices: everything up to the end of $D - 1$, which the auction on $D - 2$ published. Wind, solar and load: actuals up to the cut-off minus one hour of publication lag. Every model receives the last 28 days of 15-minute history (the context). A model may train on longer history provided each training row is built with exactly the history that was public at that row's own cut-off, and training never happens inside the live window: models that retrain do so in the nightly catch-up at 03:30, and a live run that finds no trained model falls back to the latest earlier one rather than train. The TSOs' day-ahead renewable forecasts are not an input: Regulation (EU) 543/2013 [15] has them published by 18:00 on $D - 1$, after the gate.

3.3 Actuals and provenance

The primary source is the ENTSO-E Transparency Platform [16], parsed from its XML: day-ahead prices (document A44, the 15-minute SDAC prices), actual generation per production type (A75; wind as onshore B19 plus offshore B18, solar B16) and actual total load (A65). ENTSO-E does not deliver every quarter-hour: alone it lacked 12% of Belgian and 2.6% of French wind quarter-hours in September 2026. Gaps are filled from the TSOs' own open data (Elia for Belgium, RTE éCO2mix for France), then from Energy-Charts as a last resort; where a TSO publishes finer data than it sends ENTSO-E, the TSO comes first (Elia sends hourly wind and solar to ENTSO-E but publishes quarter-hours). Every snapshot and every set of actuals records how many points came from each source, and the site shows it. Electricity Maps' choice of sources served as the reference for the new zones, with one exception found in operation: ENTSO-E's DK1 offshore wind covers only the parks registered there and has missed 0.3–0.5 GW of new offshore capacity since July 2026, so Danish wind has come from Energinet's real-time production data since delivery day 18 October (Danish solar and load stay on ENTSO-E). The first two live issue days, 2 and 3 October, were run and scored on Energy-Charts data alone, with wind and solar scored on $D + 2$.

The Netherlands shows why provenance matters. ENTSO-E's Dutch series cover only what TenneT meters itself: solar at about 1% of the fleet (a capacity factor of 0.1% over 20 September–7 October 2026, against 10.7% in the national estimate, 12.7% in Belgium and 13.6% in France), wind mostly offshore, and load net of embedded solar (about 6 GW against 17 GW gross at a sunny midday). The Dutch wind, solar and load targets therefore come exclusively from NED, the Nationaal Energie Dashboard backed by TenneT and the distribution operators, which publishes national estimates: wind as onshore plus metered offshore, load as gross consumption as Elia and RTE report it. These series take no fallback, since a gap is better than a mixture of definitions. Live forecasts made from the earlier ENTSO-E definition (issue days up to 10 October for wind and solar, 11 October for load) are kept and shown but never scored. When a series' definition changes but its earlier scores remain valid, the earlier actuals are locked: NED's offshore wind was a park-by-park model through delivery day 12 October, 7–27% above the metered output used since, and those days keep their scores.

Actuals are *frozen* the first time they are complete and scored from the frozen copy. Until $D+3$, wind, solar and load are re-fetched at every scoring run; a change above 0.5% of the day's energy re-freezes the actuals and re-scores the day, and every check is logged. Prices are final at the auction. Between the gate and scoring, the site shows the quarter-hours metered so far as provisional actuals, which are never scored.

3.4 Covariates frozen at the cut-off

Models that declare covariate inputs receive, per issue day, a table frozen at the cut-off and a log of what went into it.

Weather. Three independent global and regional NWP models, ICON-EU (DWD), GFS (NOAA) and IFS HRES (ECMWF), are fetched run by run from the Open-Meteo single-runs archive [17]. A run is admitted only if its initialisation is at most the cut-off minus a lag, the provider's measured publication delay plus a margin (3.5 h for ICON-EU, 7 h for GFS and IFS); at 11:30 this admits the 06 UTC ICON-EU run and the 00 UTC GFS and IFS runs. Since 19 October a run must also be complete: if any variable is less than 99% filled over lead times 12–48 h, which contain the delivery day, the fetch is retried up to five times and the builder then falls back to an earlier cycle, at most two (Section 5). Weather is sampled at 153 capacity-weighted k-means centroids in all, per zone 8 for onshore wind, 8 for solar, 6 for load (weighted by population) and 2–4 for offshore wind. They are built from public data: OpenStreetMap turbine positions scaled to Elia's capacity and municipal solar data of the distribution operators for BE, RIVM turbine positions and CBS municipal solar capacity for NL, ODRÉ's national register of installations at commune level for FR, the Marktstammdatenregister for DE-LU and Energinet's capacity per municipality for DK1 and DK2. The table carries, per variable and bucket, each NWP model's capacity-weighted zone mean, and across the three models the ensemble mean, spread and pairwise

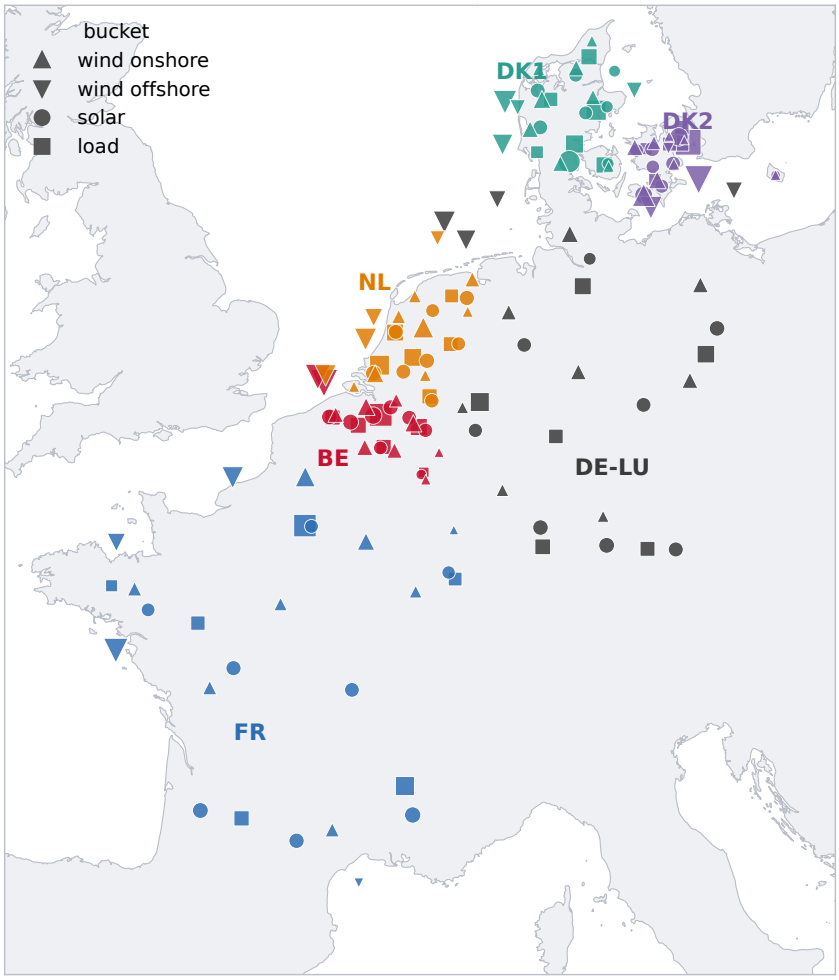


Figure 2: The 153 weather sampling points: capacity-weighted k-means centroids per zone and bucket, from public registers (load: population). Marker size is the point's share of its zone and bucket.

differences, so that disagreement can serve as a feature. Models trained on the 28-day context train on tables built by the same rule day by day, from the runs that were available at each day's own cut-off. The two-year training frames of `lgbm_prod` and `lin_cov_2y` predate the single-runs archive for ICON-EU and GFS; there the archive's previous-runs product is matched to the live lead time to the nearest day, which gives forecasts of the same lead but not always of the same run (Section 5). Figure 2 shows the sampling points.

Load forecast. ENTSO-E's day-ahead total load forecast (A65), which TSOs must publish two hours before the gate. It is an input of the price models; load models also received it until issue day 15 October and no longer do, because in most zones it is itself the pre-gate reference they are compared with (Section 4.7).

Fuel cost. Daily closes of TTF gas (ICE Dutch front month) and of an exchange-traded EU carbon allowance certificate, as published by Yahoo Finance, the last close dated two days before delivery or earlier, and the derived short-run marginal cost of a gas plant, $\mathrm{TTF}/0.52 + 0.37\,\mathrm{EUA}$ (EUR/MWh).

3.5 Entries

A model is a folder with a declarative `model.yaml`: a named sktime [18] estimator, a Python module, or a subprocess script with its own environment, the route all foundation models take (each model loads its weights once per run, for all series). A model whose environment is missing is skipped and logged, not failed. The live run takes 10–20 minutes for the field on a four-core, 8 GB server (17 minutes for 24 series on 15 October), against a 30-minute window; from 16 October two models run concurrently, each with half the CPU threads. Dropping a folder into `models/` is the whole act of deployment: the live run picks up every enabled folder, and the nightly catch-up runs any new model over a fixed backtest window (14 September – 1 October 2026, shown on the site as a separate record and not used in this paper). The field on 27 October (Table 2) has 42 active entries.

No TSO in France or Germany publishes a day-ahead wind or solar forecast before the gate (RTE at 16:15, the German TSOs at 18:00 on $D-1$), and Energinet's is not verified as published before it, so `tso_cutoff` has wind and solar only for BE and NL. Classical univariate models (ARIMA, ETS, Theta, CES, TBATS) were in the field in its first two weeks and were retired: on 15-minute data with a 96-step daily cycle and a 28-day context they were not competitive. Their records are kept but no longer shown.

3.6 Scoring

For every forecast, issue day d and series s the benchmark records MAE, RMSE, bias and, for models with deciles, the weighted interval score, the Winkler score and coverage of the 80% interval, and the pinball loss averaged over the nine deciles,

$$\mathrm{PL} = \frac{1}{9\,T}\sum_{t=1}^{T}\sum_{\tau\in\{0.1,\ldots,0.9\}} \max\big(\tau\,(y_t - q_{\tau,t}),\ (\tau-1)(y_t - q_{\tau,t})\big), \tag{1}$$

with T the scored quarter-hours of the day. Errors are compared across series of different scale by relating them to the baseline `naive_weekly`, and they are aggregated *before* they are normalised: for model m, series s and the set $\mathcal{D}_{m,s}$ of issue days on which both m and the baseline were scored,

$$\mathrm{RelMAE}_{m,s} = \frac{\sum_{d\in\mathcal{D}_{m,s}}\sum_t |e^{m}_{d,t}|}{\sum_{d\in\mathcal{D}_{m,s}}\sum_t |e^{\mathrm{base}}_{d,t}|}, \qquad \mathrm{RelRMSE}_{m,s} = \Big(\frac{\sum_{d,t}(e^{m}_{d,t})^2}{\sum_{d,t}(e^{\mathrm{base}}_{d,t})^2}\Big)^{1/2}. \tag{2}$$

A ratio of means rather than a mean of daily ratios keeps one calm day, on which the baseline happened to be nearly right, from dominating the average. Over several series the per-series values are combined by the mean, the median or the geometric mean; the site lets the reader switch. The geometric mean, used in the tables below, treats halving and doubling the error symmetrically and penalises a model that fails on one series.

Three choices in these definitions deserve to be explicit, because each changes what "best" means. *Which summary of the predictive distribution is rewarded.* Absolute error is minimised by the median of the predictive distribution and squared error by its mean [11]; for skewed targets such as solar power or spiking prices the two differ, and a model trained on quantile loss is not wrong to forecast the median. The field mixes both: most foundation models report the median of their predictive quantiles (Chronos, TiRex, Moirai, Sundial, Toto, t0, Timestral, Tabby), while TimesFM and the LightGBM and ridge entries, fitted on squared error, report a conditional mean, as does the climatology. The standings are built on MAE and so reward the median; Rel. RMSE is shown next to it so that mean-oriented entries are not judged on the other functional only, and bias is reported separately because neither accuracy measure distinguishes a systematic offset from noise. *Which benchmark is in the denominator.* Every scale-free error embeds a benchmark [12]. Energy Bench names it: the same quarter-hour one week earlier, a seasonal naive forecast that respects the daily and weekly cycles of price and load and needs no fitting, evaluated on the same test days as the model rather than in-sample, which is possible because every series has 96 scored values per day [13]. For wind and solar it is a weak benchmark, and relative errors well below one are expected there; the benchmark is a common unit, not a claim about what is hard.

Table 2: The field, 27 October 2026 (42 active entries). Parameter counts are those of the checkpoints the benchmark loads. Covariates: *native*, the model's own known-future covariate input; *ridge*, an in-context ridge regression on the covariates whose residual the model forecasts; *full*, the whole covariate table.

Entry	Params	Covariates	Developer, set-up
Baselines			
`naive_weekly`			same quarter-hour one week earlier; the reference of all relative errors
`naive_daily`			same quarter-hour one day earlier
`climatology`			same local time of day within ± 14 days, 2023 to August 2026
`lin_lags`, `lgbm_lags`			ridge and LightGBM on the series' own lags; two years, retrained monthly
Models trained on the covariates[a]			
`lin_cov`, `lgbm_cov`		full	ridge and LightGBM on the 28-day context
`lin_cov_2y`		full	ridge on two years
`lgbm_prod`		full	LightGBM on two years, 120-day recency half-life, solar on its capacity factor[b]
Foundation models, zero-shot[c]			
Chronos-2 [19]	120M	native	Amazon
Chronos-Bolt base [19]	205M		Amazon
TimesFM 2.5 [20]	231M	ridge	Google
TimesFM 3.0 [20]	331M	native	Google
Moirai 2 small [21]	11M	[d]	Salesforce
TiRex [22]	35M		NX-AI
TiRex-2 [22]	95M	native	NX-AI; about 38M active in univariate mode
Toto	151M	[d]	Datadog
Toto 2	313M, 1B		Datadog; live from 25 October
Sundial	128M		Tsinghua University
TTM r2	3.4M		IBM
Tabby, Tabby-Pretrain	147M, 146M	ridge	Huawei Noah's Ark Lab; load forecast without covariates
Timestral	209M	native	Mistral AI
t0-alpha, t0-beta, t0-gamma	102M, 256M, 256M	native	The Forecasting Company
Ensembles: quarter-hourly median of fixed members, point and each quantile; live from 20 October			
`ens_fm`			the newest history-only model of the five native-covariate families
`ens_fm_cov`		native	their five covariate variants
`ens_hybrid`		native	those five and `lgbm_prod`
References: the grid operators' forecasts			
`tso_cutoff`			as published before the gate; ranked; live only[e]
`tso_final`			final day-ahead wind and solar, published after the gate; never ranked[f]

[a] *Full*: every NWP model's columns, the ensemble statistics and differences, the calendar and, for price, yesterday's price. The foundation models take a compact subset instead: three or four ensemble means and spreads per target, and for price the load forecast, fuel cost and the renewable drivers of the coupled zones.

[b] Retrained daily for price and load, monthly for wind and solar.

[c] Every foundation model runs on the series' own history; those with a covariate entry run a second time with the frozen covariates (e.g. `chronos2` and `chronos2_cov`).

[d] Covariate variants retired after trial runs: covariates made Moirai 2 worse, and Toto gained too little for its run time and memory.

[e] Elia's 11:00 day-ahead wind and solar forecast for BE; NED's national wind, solar and load forecast for NL, fetched at the cut-off; the ENTSO-E day-ahead load forecast elsewhere.

[f] Elia 18:00, RTE 16:15, ENTSO-E for DE-LU and Danish wind, NED in the evening for NL.

How the dimensions are aggregated. Errors are averaged over the 96 quarter-hours of the delivery day and over issue days within a series, then across series with equal weight per series. Equal weight is a choice: it asks which model forecasts best across forecasting problems, not where the most megawatts or euros are; the forecast-value view (Section 3.8) gives the latter answer for wind and solar.

3.7 Ranking

A *race* is one issue day and one series; its field is every model with a scored forecast in it. A model whose run failed is left out of that race rather than ranked last (the run tracker on the site shows every failure), and a model joins the races of the series it declares from the day it is added. Within a race every pair of models plays one game, won by the lower relative error of the day (equal errors split the game). All games of the selected period are fitted at once with a Bradley–Terry model [23], $P(i \text{ beats } j) = p_i/(p_i + p_j)$, by the minorisation–maximisation algorithm of Hunter [24], and reported on the Elo scale, $\mathrm{Elo}_i = 1000 + 400 \log_{10} p_i$, with the geometric mean of the strengths fixed at 1 so that 1000 is the field's average. One virtual drawn game against an average opponent keeps the rating finite for a model that wins or loses every game. Unlike a running Elo, the rating does not depend on the order of the races, and a win over a strong model counts more than a win over a weak one; the same construction underlies the Chatbot Arena leaderboard [25].

The 95% interval is the 2.5th–97.5th percentile of 200 refits on issue days drawn with replacement. Days rather than races are resampled because all races of a day share the same weather and tend to be won and lost together. Overlapping intervals mean the order of two models is not settled. Any pairwise rating is relative to the field: a model's score depends on whom it played, and several near-identical entries (three releases of one model family, say) take wins from each other and from everyone else. The pooled relative errors do not depend on the field and are always reported next to the rating. A model is listed as provisional if it has fewer races than the smaller of 24 and half the most any model has. The standings can be restricted to the last 7 or 30 days, to a zone or a target, and to models with or without covariates; filters choose which races count and which models are shown, never how a race is scored.

3.8 Forecast value

Accuracy is not revenue. For wind and solar, the benchmark also values each forecast as the day-ahead income of a price-taking producer that owns a virtual 100 MW asset in each zone, producing like the zone's whole fleet: actual output and every forecast are scaled by 100 MW over the installed capacity, so that every zone counts equally. Before the gate the producer sells a fixed percentile of the forecast distribution (or the point forecast) at the day-ahead price. Each deviation is settled at the zone's published imbalance price (ENTSO-E document 17.1.G; the German price is the national reBAP). By default only the error cost counts: a deviation of size δ costs $|\delta| \cdot |\pi^{\mathrm{imb}} - \pi^{\mathrm{DA}}|$, with the imbalance price of its own direction, so no error ever earns and the score, revenue as a share of perfect-foresight revenue, is at most 100%. The alternative settles deviations as each zone would pay them, two prices where the TSO publishes two (NL and FR) and one elsewhere; there a deviation can earn when the imbalance price happens to sit on its side, which over a few weeks rests on a handful of price spikes. The producer cannot curtail and does not trade intraday.

3.9 Operation and reproducibility

The benchmark is a set of systemd timers on one server: the live run at 11:30, scoring at 07:00, 14:45 and 20:45, an hourly fetch of provisional actuals, and a catch-up at 03:30. Every forecast, input snapshot, covariate table, provenance file and revision log is kept, and the website is a static export of them, so any number on it can be recomputed. The configuration file that defines the benchmark (cut-off, gate, resolution, deciles, context, baseline, backtest window, sources, admission lags) is versioned with the code: changing it changes the benchmark. On the site, backtest and live results are always shown apart; this paper reports live results only.

Table 3: Scored issue days per series in the live record, 2–25 October 2026.

	BE	NL	FR	DE-LU	DK1	DK2
Price	24	24	24	11	11	11
Wind	24	15	24	11	8	8
Solar	24	15	24	11	11	11
Load	20	14	20	11	–	–

4 Results: the first twenty-four live days

The results below are those of the live record only, scored on the forecasts as they were locked before each gate. After the coverage of the record, they give the standings, the value of the covariates for foundation models, the role of model size, and the comparison with purpose-built models, with the full field over its last six days and with the grid operators' own forecasts.

4.1 Coverage

The live record covers issue days 2–25 October 2026, the last day with every target scored when this paper was written. It is uneven by design: series enter the record when they can be scored on a stable definition, and not before (Table 3). BE, NL and FR price, wind and solar run from 2 October and load from 6 October; the Dutch wind, solar and load forecasts made before the switch to NED's national series (Section 3.3) are voided, so the Dutch record starts on 11–12 October. DE-LU, DK1 and DK2 run from 15 October. Danish wind lost two issue days to its own switch of source. Danish load has been published by ENTSO-E only in part since 10 October, which leaves four scored days and a pre-gate reference that does not match them; it is left out. The record therefore holds 22 of the 24 series, with between 8 and 24 issue days each. Models also joined at different times, which is why every comparison below is made on the races the compared models share.

4.2 Standings

Table 4 gives the standings for the 39 entries with at least 100 races and for the pre-gate TSO reference, and Figure 3 shows their ratings. The two median ensembles of covariate models, live since 20 October, rate highest (`ens_hybrid`, Elo 1265 and Rel. MAE 0.278; `ens_fm_cov`, 1232 and 0.288). Among single models, t0-gamma, TimesFM 3.0, Chronos-2 and Timestral with covariates lead (Elo 1193–1212, Rel. MAE 0.320–0.335); their intervals overlap each other's, the ensembles' and that of the grid operators' pre-gate forecast (1185). Behind the leading group the field splits cleanly in two. Every covariate-aware foundation model is separated by its interval from every history-only model, and the best history-only entries, the history-only ensemble and Timestral (Elo 973 and 968), sit behind fifteen covariate models. Among the history-only foundation models the differences are small (pooled Rel. MAE 0.62–0.76) next to the effect of giving any of them the weather (0.32–0.41 with covariates, Table 5).

Figure 4 breaks the pooled error down by series. The covariate models lead on wind and solar in every zone, by a margin that dwarfs the differences between them; Dutch solar is the hardest of these series. On load the history-only foundation models are level with the covariate ones, and the grid operators' pre-gate load forecast is strong only in France.

4.3 Covariates halve the error of foundation models

Ten foundation models run both on the series' own history and with the frozen covariate table, so the value of the weather can be read off model by model on exactly the same races (Table 5). For the seven that take covariates natively, the covariates cut the pooled relative error by 45–50%: by 58–69% on wind and 58–63% on solar, where a model that sees only the last four weeks of output cannot know tomorrow's wind or cloud; by 17–38% on price, through the load forecast, fuel cost and the expected

Table 4: Live standings, issue days 2–25 October 2026, all six zones (22 series). Elo with its day-bootstrap 95% interval; Rel. MAE and Rel. RMSE pooled per series against `naive_weekly`, geometric mean over the series a model forecasts ("all") or over the zones of one target. Entries with at least 100 races; covariate models in bold.

	Model	Elo	95% interval	Races	Rel. MAE	Rel. RMSE	Price	Wind	Solar	Load
					all series		Rel. MAE per target			
1	**ens_hybrid**	1265	[1205, 1317]	130	0.278	0.311	0.271	0.199	0.270	0.494
2	**ens_fm_cov**	1232	[1176, 1280]	130	0.288	0.323	0.276	0.211	0.280	0.508
3	**t0_gamma_cov**	1212	[1164, 1257]	214	0.320	0.350	0.303	0.250	0.292	0.575
4	**timesfm3_cov**	1201	[1171, 1226]	349	0.321	0.343	0.290	0.223	0.315	0.663
5	**chronos2_cov**	1197	[1159, 1233]	349	0.335	0.366	0.293	0.249	0.334	0.641
6	**timestral_cov**	1193	[1162, 1229]	214	0.324	0.352	0.348	0.229	0.283	0.595
7	tso_cutoff	1185	[1125, 1246]	137	0.400	0.414	–	0.191	0.366	0.606
8	**t0_beta_cov**	1148	[1127, 1165]	335	0.361	0.392	0.325	0.272	0.363	0.643
9	**tabby_cov**	1130	[1060, 1203]	214	0.369	0.387	0.351	0.279	0.356	0.632
10	**tabby_pretrain_cov**	1127	[1072, 1181]	214	0.368	0.387	0.350	0.282	0.346	0.652
11	**lgbm_prod**	1122	[1078, 1164]	214	0.351	0.358	0.347	0.216	0.364	0.698
12	**tirex2_cov**	1122	[1095, 1145]	310	0.350	0.367	0.349	0.220	0.348	0.715
13	**t0_alpha_cov**	1101	[1082, 1126]	335	0.377	0.398	0.417	0.230	0.347	0.767
14	**lgbm_cov**	1080	[1048, 1116]	349	0.380	0.392	0.387	0.209	0.349	1.031
15	**timesfm25_cov**	1075	[1037, 1111]	349	0.408	0.429	0.382	0.256	0.382	0.995
16	**lin_cov_2y**	1018	[986, 1050]	248	0.458	0.467	0.340	0.268	0.451	1.638
17	ens_fm	973	[934, 1004]	130	0.569	0.597	0.514	0.611	0.621	0.524
18	timestral	968	[931, 998]	214	0.621	0.647	0.494	0.731	0.680	0.600
19	t0_gamma	958	[923, 992]	214	0.626	0.647	0.489	0.668	0.720	0.663
20	timesfm3	957	[926, 989]	349	0.646	0.673	0.466	0.660	0.854	0.670
21	chronos2	955	[923, 984]	356	0.651	0.685	0.479	0.653	0.898	0.637
22	timesfm25	954	[918, 988]	349	0.634	0.667	0.456	0.606	0.799	0.790
23	**lin_cov**	950	[912, 978]	248	0.513	0.509	0.458	0.274	0.555	1.382
24	tabby_pretrain	946	[911, 976]	214	0.630	0.654	0.518	0.686	0.686	0.652
25	chronos_bolt	929	[891, 958]	349	0.677	0.710	0.501	0.653	0.937	0.689
26	tirex2	927	[899, 953]	310	0.661	0.681	0.497	0.652	0.868	0.686
27	moirai2	925	[877, 968]	349	0.679	0.716	0.483	0.652	0.929	0.752
28	t0_beta	925	[895, 955]	335	0.665	0.692	0.504	0.654	0.902	0.655
29	tabby	925	[876, 966]	214	0.673	0.701	0.516	0.755	0.817	0.632
30	tirex	910	[884, 933]	349	0.683	0.707	0.510	0.657	0.823	0.850
31	lin_lags	901	[871, 933]	248	0.656	0.660	0.470	0.622	0.850	0.798
32	t0_alpha	900	[866, 933]	335	0.687	0.710	0.500	0.716	0.829	0.783
33	lgbm_lags	881	[821, 925]	356	0.745	0.779	0.520	0.736	0.908	0.966
34	sundial	880	[844, 915]	349	0.718	0.721	0.539	0.729	0.938	0.722
35	toto	841	[797, 883]	349	0.758	0.790	0.525	0.708	0.960	1.020
36	ttm_r2	832	[803, 861]	349	0.762	0.743	0.535	0.656	0.900	1.262
37	naive_daily	784	[729, 830]	356	0.930	0.978	0.649	0.758	0.995	1.955
38	naive_weekly	772	[715, 817]	356	1.000	1.000	1.000	1.000	1.000	1.000
39	climatology	759	[718, 788]	349	1.033	0.993	1.171	0.743	0.847	1.889
–	tso_final (ref.)	–	–	–	0.235	0.249	–	0.219	0.260	–

Races differ because entries joined at different times (ensembles 20 October; t0-gamma, Timestral, Tabby, `lgbm_prod` 16 October); Table 6 compares the whole field on common days. `tso_cutoff` forecasts 8 of the 22 series (BE and NL wind and solar; load in BE, NL, FR and DE-LU). `tso_final` covers wind and solar in BE, NL, FR and DE-LU and wind in DK1 and DK2; its Danish forecasts cover a different scope from the actuals (Section 4.7). Toto 2 (313M and 1B), live from 25 October, has too few races to list.

renewable infeed; and on load hardly at all (-13% to $+4\%$), since a load curve is dominated by daily and weekly cycles that the history already shows. Of the three that receive the covariates through an in-context ridge regression, the two Tabby models gain nearly as much (42–45%), while TimesFM 2.5 gains less (36%), little on price, and its load error rises by 26%, where the ridge fitted on 28 days extrapolates badly.

The effect does not depend on the zone (Figure 5, where every point of a zone is computed on the same races). In each of the six zones the covariates cut Chronos-2's error by 38–69% on wind and solar and by 32–46% on price, while on load they change it by between -4% and $+10\%$. On wind and solar the grid operators' forecasts are ahead of Chronos-2 with covariates in every zone where they exist,

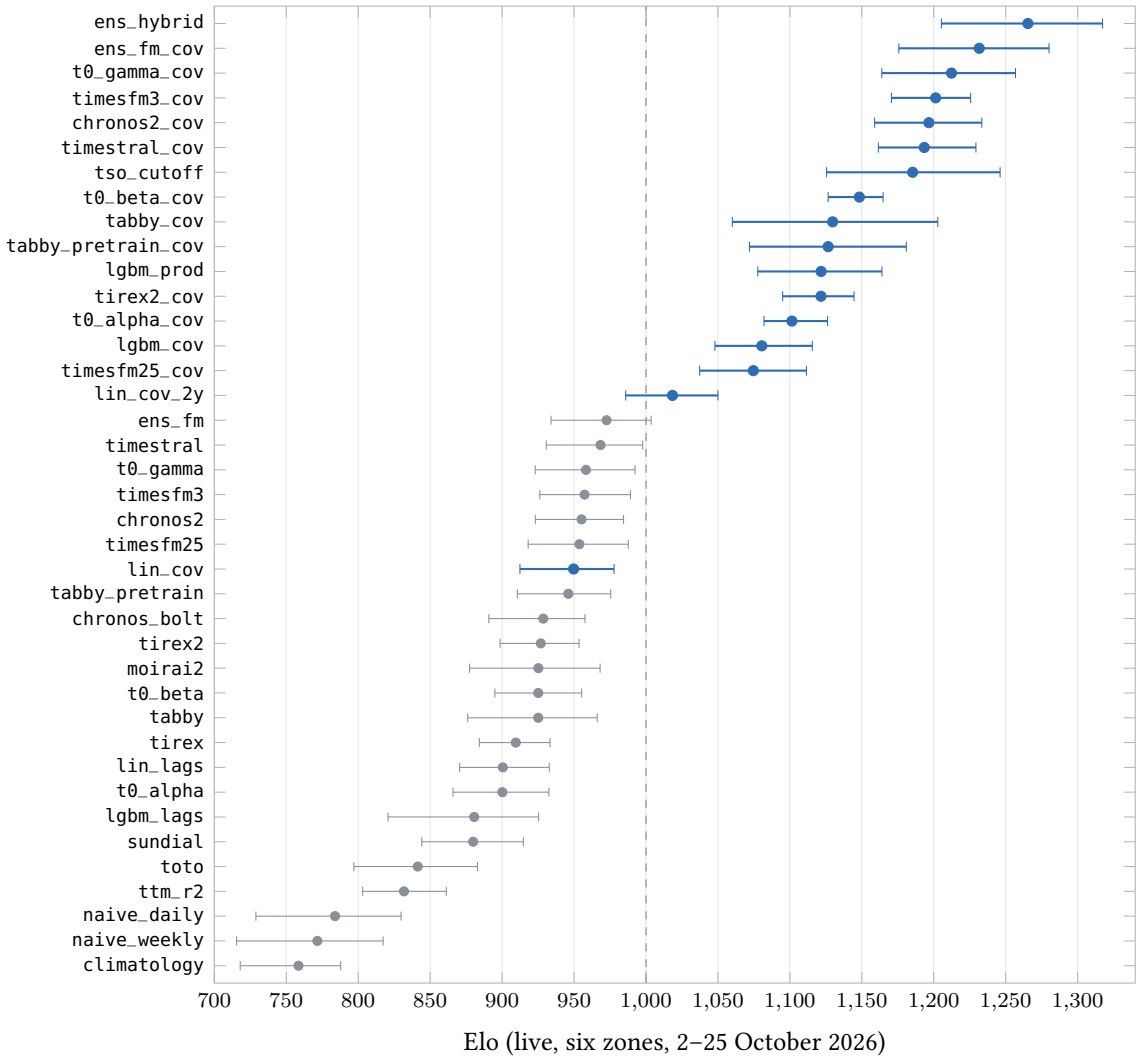


Figure 3: Live Elo ratings with day-bootstrap 95% intervals, entries of Table 4. Blue: models using the frozen covariates, and the pre-gate TSO forecast `tso_cutoff`; grey: history only. The dashed line is the field average.

and `lgbm_cov` beats both of them in NL; on load, Chronos-2 with covariates is ahead of the pre-gate forecast in BE, NL and DE-LU and behind it in FR.

Figure 6 shows what the difference looks like on two delivery days on which the covariates helped more than in the median race; in the median wind or solar race, Chronos-2 with covariates has 41% of the error of Chronos-2 without, on these two days 21% and 28%. On 8 October Belgian wind blew at 3.3–3.9 GW through the morning and fell to about 1.1 GW by the evening. Chronos-2 on its own history carried its recent output forward and forecast less than 0.3 GW all day; the covariate version followed the forecast weather, though it was late on the evening decline, as was Elia's pre-gate forecast. On 14 October French solar peaked at 15.8 GW: Chronos-2 alone reached 10.6 GW, the covariate version 13.8 GW, and RTE's final forecast, published after the gate, 15.7 GW.

4.4 Model size

Figure 7 sets each foundation model's error against its number of parameters, on the ten issue days 16–25 October, on which all sixteen ran (214 races each). Without covariates, larger models do somewhat better (Spearman $\rho = -0.58$, $p = 0.02$), but the spread is small and not monotone: the best five, from 120M (Chronos-2) to 256M (t0-gamma), are within 0.01 of each other, and Toto, at 151M, is the weakest,

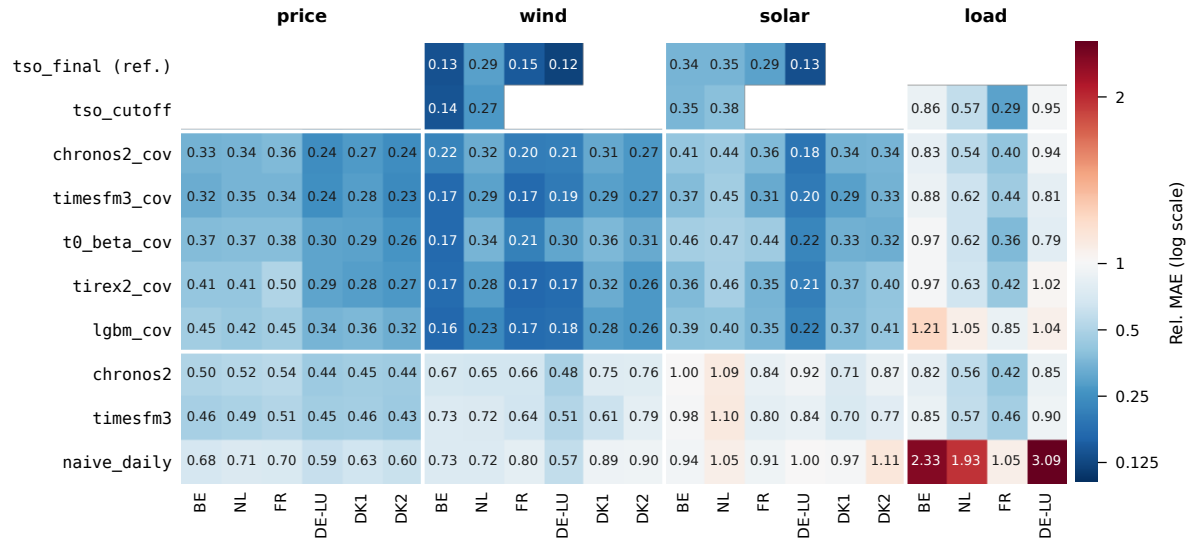


Figure 4: Pooled Rel. MAE per series for the references, five covariate models, two history-only foundation models and `naive_daily` (live, 2–25 October 2026; colour on a log scale, 1 is the seasonal-naive baseline). Empty cells: no forecast, or (for `tso_final` in DK1 and DK2) a forecast of a different scope.

Table 5: The same foundation model without and with the frozen covariates, on the races both forecast (live, all zones, issue days 2–25 October 2026). "Native": the model takes known-future covariates itself; "ridge": an in-context ridge regression on the covariates, the model forecasting its residual (Tabby forecasts load without covariates). Rel. MAE pooled per series, geometric mean over series; a change is the geometric mean of the per-series ratios minus one.

				Rel. MAE		Change with covariates				
Model	Covariates	Days	Races	without	with	all	price	wind	solar	load
Chronos-2	native	23	349	0.650	0.335	−48%	−38%	−62%	−63%	+1%
TimesFM 3.0	native	23	349	0.646	0.321	−50%	−38%	−66%	−63%	−1%
t0-alpha	native	21	335	0.687	0.377	−45%	−17%	−68%	−58%	−2%
t0-beta	native	21	335	0.665	0.361	−46%	−36%	−58%	−60%	−2%
t0-gamma	native	10	214	0.626	0.320	−49%	−38%	−63%	−59%	−13%
TiRex-2	native	18	310	0.661	0.350	−47%	−30%	−66%	−60%	+4%
Timestral	native	10	214	0.621	0.324	−48%	−29%	−69%	−58%	−1%
TimesFM 2.5	ridge	23	349	0.634	0.408	−36%	−16%	−58%	−52%	+26%
Tabby	ridge	10	214	0.673	0.369	−45%	−32%	−63%	−56%	+0%
Tabby-Pretrain	ridge	10	214	0.630	0.368	−42%	−33%	−59%	−50%	+0%

behind TTM r2 at 3.4M. From about 100M up the relation is not significant ($\rho = -0.45$, 13 models), nor is it with covariates ($\rho = -0.41$, ten models), where TimesFM 3.0 (331M) and t0-gamma (256M) lead Chronos-2 (120M) by 0.02. Every covariate model, whatever its size, beats every history-only model: the weather is worth far more than any difference in size.

4.5 Against purpose-built models

The covariate-aware foundation models are used zero-shot, yet the five that take covariates natively and have run over the whole record all beat a LightGBM trained on the same covariate table (`lgbm_cov`, Rel. MAE 0.380, against 0.321–0.377), although LightGBM receives more of it: every NWP model's columns, the calendar and, for price, yesterday's price, against a handful of ensemble means and spreads. LightGBM is the best of them on wind (0.209, against 0.220–0.272) and level on solar (0.349, against

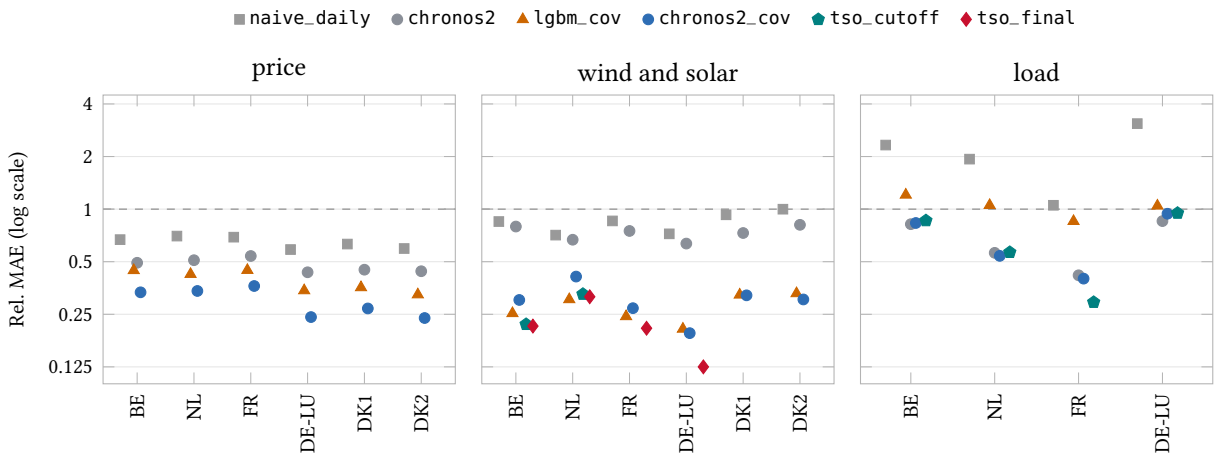


Figure 5: Pooled Rel. MAE per zone and target group (live, 2–25 October 2026, log scale; the dashed line is the seasonal-naive baseline). Every point of a zone and panel is computed on the same races: those that all models shown there forecast, the TSO references included where they exist (`tso_cutoff`: BE and NL wind and solar, and load; `tso_final`: wind and solar outside Denmark, Section 4.7). Danish load is left out (Section 4).

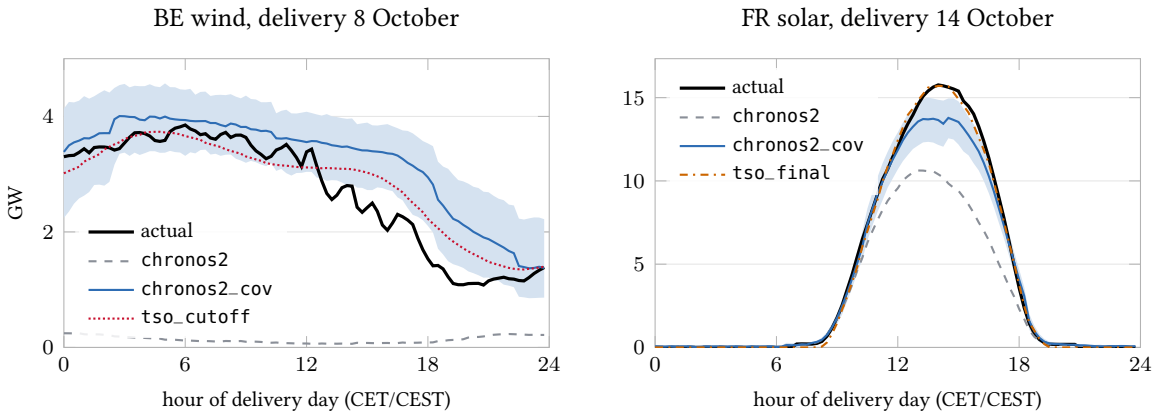


Figure 6: Two delivery days: actual output, Chronos-2 without and with covariates (shaded: the covariate version's 10–90% band), and a TSO forecast (Elia's pre-gate forecast for BE; RTE's final forecast, published after the gate, for FR).

0.315–0.363), but far behind on load (1.031), where 28 days are too few to learn the weather response. Trained on two years, `lgbm_prod` does better: on the days both ran (16–25 October) it reaches 0.351 against 0.377 for `lgbm_cov`, level with the native-covariate foundation models in the middle of the field.

4.6 The full field over the last six days

Nine entries joined on 16 October (t0-gamma, Timestral, Tabby and Tabby-Pretrain, each with and without covariates, and `lgbm_prod`) and the three ensembles on 20 October. Table 6 compares the whole field on the six issue days 20–25 October, on which every entry ran (130 races each). The median of the five covariate foundation models (`ens_fm_cov`, Rel. MAE 0.288) equals its best member, TimesFM 3.0 with covariates, and beats the other four (0.295–0.331); adding `lgbm_prod`, a model of a different class trained on the same weather, as a sixth member (`ens_hybrid`, 0.278) beats every single model. An ensemble is fixed in advance and does not need to know which member will be best, which is what makes it the more useful choice; six days are too few to rank it reliably.

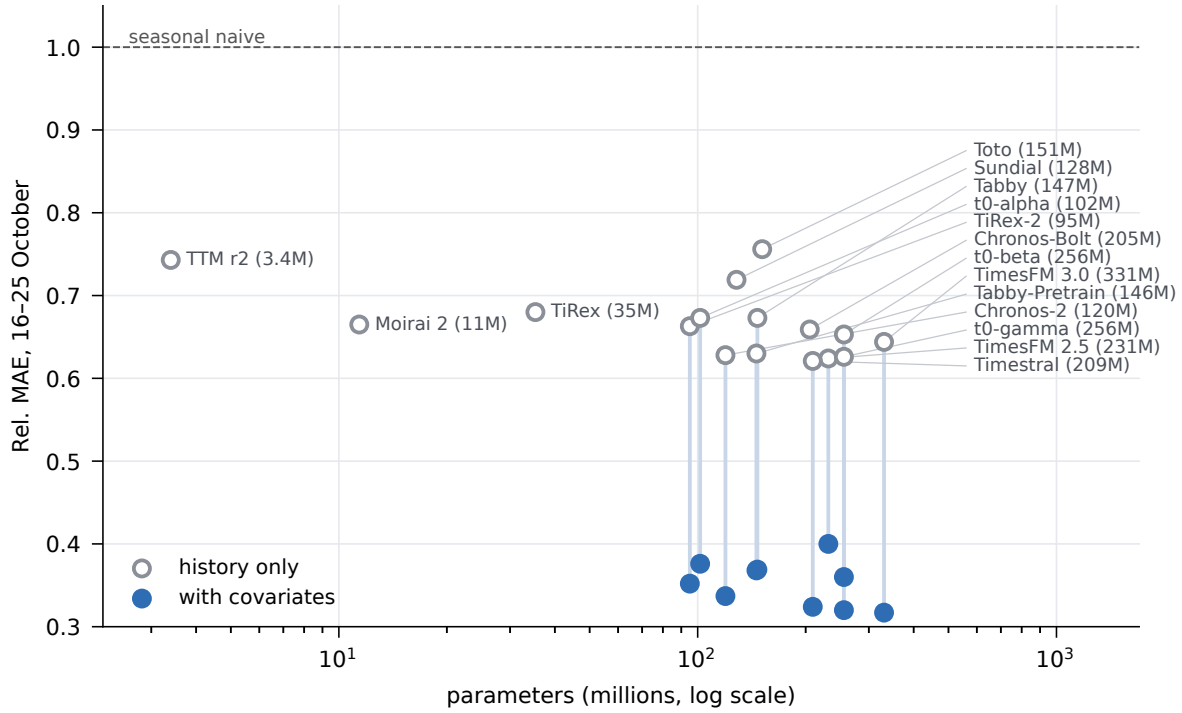


Figure 7: Pooled Rel. MAE against parameter count for the sixteen foundation models, without covariates (open) and, where a variant exists, with the frozen covariates (filled, joined to the same model without), on the issue days 16–25 October 2026 (all zones, 214 races per model). Parameter counts are those of the checkpoints the benchmark loads; TiRex-2 uses about 38M of its 95M in univariate mode.

4.7 Against the grid operators

The grid operators' forecasts are compared with each entry on exactly the races both forecast, and they remain the bar. Against the forecasts as published before the gate (`tso_cutoff`: BE and NL wind and solar, 36 races each; load in BE, NL, FR and DE-LU, 65 races), the best entries are level on wind (Rel. MAE 0.193 for `lgbm_cov`, against 0.191) but behind on solar (0.396, against 0.366) and load (0.637 for Chronos-2 without covariates and 0.641 with, against 0.606). Outside NL the pre-gate load reference is the ENTSO-E load forecast that the load models received as an input until 15 October. The final forecasts, published hours after the gate, are further ahead: on wind and solar in BE, NL, FR and DE-LU (66 races each), they reach 0.160 and 0.261, against 0.187 and 0.324 for the best entries. Denmark is left out of that comparison because its final forecast and its actuals do not measure the same thing: Energinet's day-ahead forecast for DK1 wind has run at about 0.8 of measured output for two years, a gap of level rather than of skill.

5 Limitations

The record is short and covers one season; the standings will move. The weather inputs are three deterministic models served by one provider, and their spread stands in for forecast uncertainty. That provider is a single point of failure: on six of the twenty-four live days (3, 4, 5, 10, 13 and 17 October) it served the 06 UTC ICON-EU run at the cut-off before the run was fully ingested, with missing values over the delivery day. Models built on the three-model ensemble means were barely affected; models that read each NWP model's own columns lost accuracy on those days. A completeness check has rejected such runs since 19 October; the affected days are scored as they were run. The two-year training frames match the lead time of the live runs but take ICON-EU and GFS from a previous-runs product

Table 6: The full field on the issue days 20–25 October 2026, all zones (130 races per entry; `tso_cutoff` 48). Top 16 by Elo and, for reference, the two best history-only models of Table 4 and the baseline. Columns as in Table 4.

	Model	Elo	95% interval	Rel. MAE	Price	Wind	Solar	Load
1	**ens_hybrid**	1268	[1207, 1324]	0.278	0.271	0.199	0.270	0.494
2	**ens_fm_cov**	1234	[1176, 1286]	0.288	0.276	0.211	0.280	0.508
3	**timesfm3_cov**	1225	[1170, 1279]	0.288	0.257	0.216	0.277	0.548
4	tso_cutoff	1223	[1128, 1348]	0.331	–	0.134	0.379	0.486
5	**t0_gamma_cov**	1218	[1149, 1284]	0.295	0.277	0.222	0.295	0.501
6	**timestral_cov**	1193	[1143, 1250]	0.301	0.342	0.202	0.274	0.515
7	**lgbm_prod**	1167	[1118, 1225]	0.296	0.286	0.171	0.317	0.638
8	**chronos2_cov**	1152	[1078, 1220]	0.331	0.308	0.269	0.299	0.579
9	**tabby_cov**	1145	[1047, 1245]	0.334	0.301	0.260	0.342	0.554
10	**tirex2_cov**	1139	[1095, 1184]	0.312	0.331	0.188	0.316	0.605
11	**tabby_pretrain_cov**	1129	[1039, 1215]	0.341	0.300	0.264	0.344	0.600
12	**t0_beta_cov**	1127	[1098, 1159]	0.342	0.311	0.281	0.337	0.546
13	**lgbm_cov**	1100	[1017, 1183]	0.336	0.355	0.176	0.302	0.955
14	**timesfm25_cov**	1093	[1017, 1163]	0.358	0.304	0.260	0.352	0.757
15	**t0_alpha_cov**	1092	[1047, 1133]	0.353	0.406	0.217	0.325	0.673
16	**lin_cov_2y**	1032	[969, 1078]	0.407	0.340	0.256	0.385	1.163
17	timesfm3	951	[893, 1008]	0.601	0.525	0.636	0.692	0.547
18	chronos2	948	[889, 1026]	0.586	0.520	0.622	0.663	0.534
19	naive_weekly	705	[620, 770]	1.000	1.000	1.000	1.000	1.000

that can contain runs initialised after the cut-off of the day they describe; this concerns the training data of two entries, never what is scored. The entries' settings were chosen before they went live, but by people who had seen the weeks before. The covariate models are deliberately simple and untuned; `lgbm_cov` and `lin_cov` train on the 28-day context for comparability with the foundation models, a known handicap that `lgbm_prod` removes. The Dutch targets changed definition during the record, and the weather points rest on public capacity registers of uneven detail (municipal data in BE, NL and DK, commune centres in FR, unit positions in DE-LU). The `tso_cutoff` reference exists only where a TSO publishes before the gate (BE and NL wind and solar, and load), and the Danish final forecasts measure a different scope from the actuals. The three zones added on 15 October have eleven issue days, and entries that joined late are compared with the rest only on the days they share; both will mean more with time. A model that fails is left out of the races it missed, so a model that fails often is flagged by the run tracker rather than penalised in the rating; this is deliberate and open to argument. The value computation assumes a price-taking producer without intraday trading or curtailment. Finally, a benchmark is only as informative as its entries.

6 Conclusion

A live, gated benchmark is the kind whose results cannot be improved by a more careful choice of backtest. Energy Bench runs one daily for the European day-ahead market, on quarter-hours, from frozen public data with recorded provenance, with ranking statistics that state their uncertainty and a value view that connects accuracy to revenue. After twenty-four live days in six zones the main finding is clear: the weather the forecaster could know at the cut-off matters more than the model that uses it. The same foundation model forecasts with about half the error once it is given that weather (for Chronos-2, in every zone), zero-shot foundation models with covariates beat a gradient-boosting model trained on the same covariate table, and a median of them is better still. Model size matters far less. The grid operators' own forecasts remain the bar: before the gate they are level with the best entries on wind and ahead on solar and load, and after it ahead on both wind and solar. The repository accepts entries by pull request; the next zones, references and value settlements will be added the same way.

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